

AGENTIC AI FOR ENGINEERING RESEARCH

Own *the work.*

How to delegate research work to an AI agent without delegating your scientific judgment.

PRESENTER

Harun Pirim

SUPPORT

Yusuf Akbulut

Connect Elicit Learn Lab

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Layers · Data · Disciplines

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Patterns · Mechanisms · Explanations

Learn.

Models · Decisions · Impact

Network science, machine learning and optimization for complex systems: power grids, protein interactions, human functioning and online discourse.

ENERGY & INFRASTRUCTURE

Resilience: predicting where networked systems fail.

BIOLOGY & HEALTH

Discovery: from molecular function to human functioning.

SOCIETY & INFORMATION

Dynamics: how opinion and information move.

12 researchers from industrial engineering, computer science and political science

ce11.org

THE QUESTION THIS HOUR ANSWERS

How do you delegate research work to an AI agent *without delegating your scientific judgment?*

THE AGENT

Reads a repository, proposes, writes and runs code, debugs, evaluates, reports.

YOU

Own the question, the plan, the evaluation, the conclusion.

At the end you sign the result. *Or you do not.*

What you will do today

01

Own it.

An interview turns a research question into a twenty-line plan. You commit it.

02

Execute.

The agent carries out the plan. You review the diff and commit.

03

Sign it.

You accept, qualify, or reject each claim the agent puts in front of you.

THE RESEARCH WORKFLOW

AI can help at almost every step of research



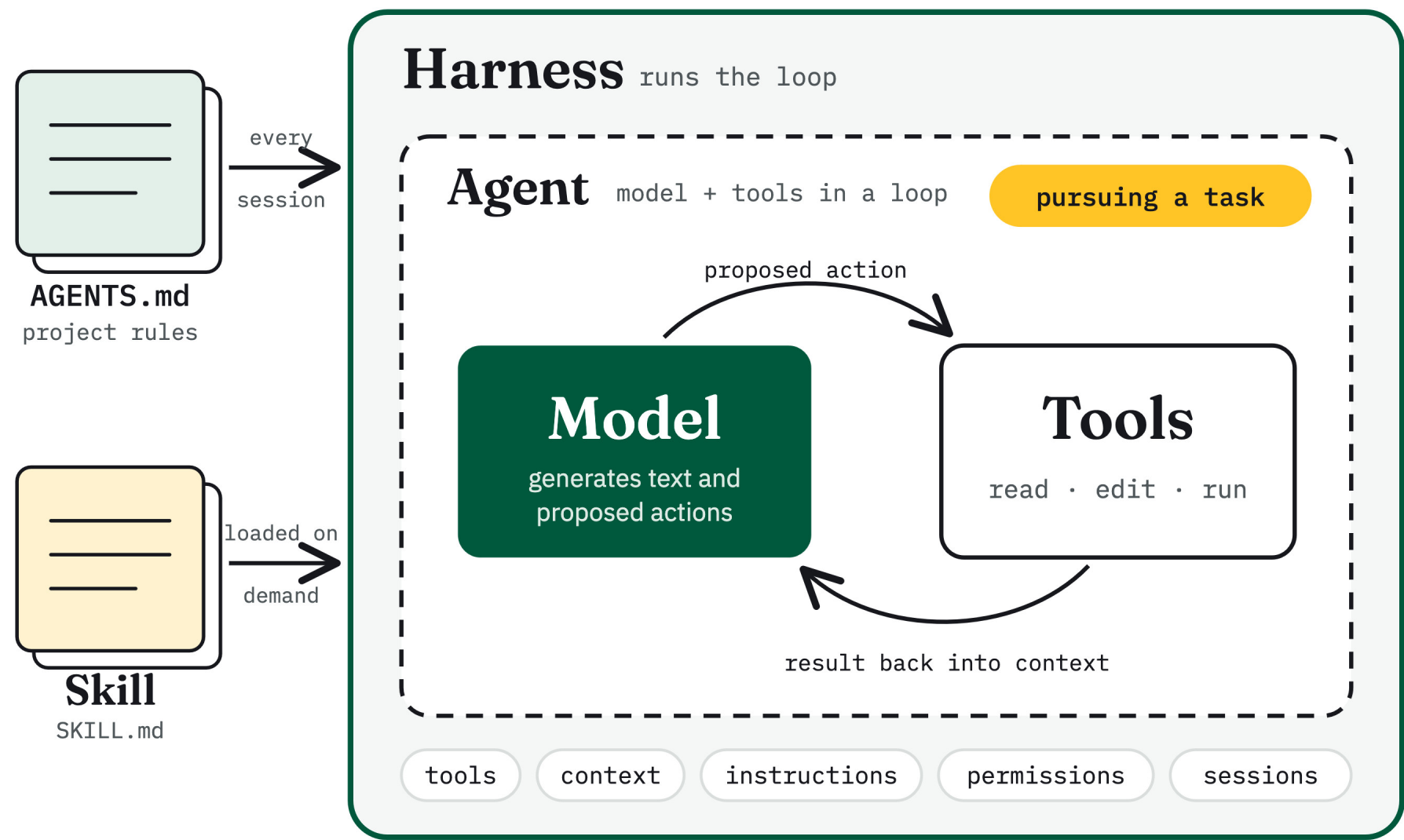
AGENTS THAT SPAN THE WHOLE LOOP

Claude Science · Anthropic’s research workbench

Biomni · Stanford’s biomedical agent

Today we focus on one step, *done properly*.

Four words and one file



Model

Generates text and proposed actions from context. (e.g. Claude, GPT, Gemini, Llama)

Agent

A model plus tools in a loop, pursuing a task. (e.g. Claude fixing a failing test: read, edit, run, repeat)

Harness

Runs the loop: tools, context, instructions, permissions, sessions. (e.g. Claude Code, Codex, pi)

Skill

A prompt in a file the harness loads on demand. (e.g. own-it, sign-it)

AGENTS.md

Standing instructions for the repository, read at the start of every session. (e.g. how to run tests, what never to touch)

Which model, how hard should it think

1 DIAL 1

Model choice

Subscription cheapest now

Flat fee. Best for daily agent work.

API

Pay per token. Adds up fast.

2 DIAL 2

Reasoning effort

More effort costs more time and money.

It does not guarantee a better answer.

3 DIAL 3

Where it runs

Hosted

Strongest. Data leaves the machine.

Self-hosted

Lab server. Data stays inside.

Local

Your laptop. Small, free, private.

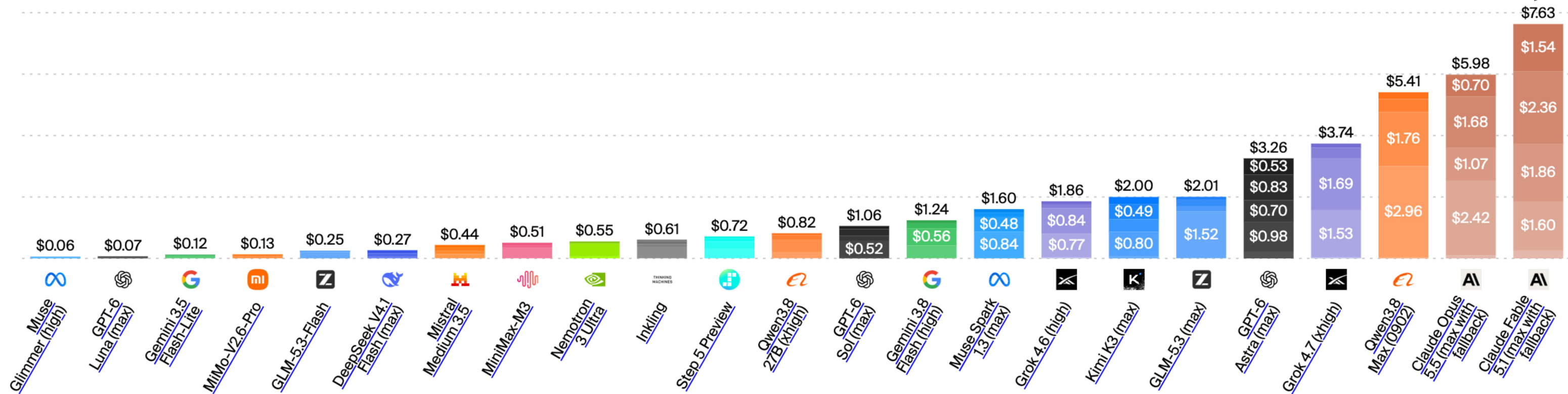
What agent work costs at API prices

Cost per Intelligence Index Task

Weighted average cost (USD) per Artificial Analysis Intelligence Index task, segmented by token type. Lower is better

● Answer ● Reasoning ● Cache Write ● Cache Hit ● Input

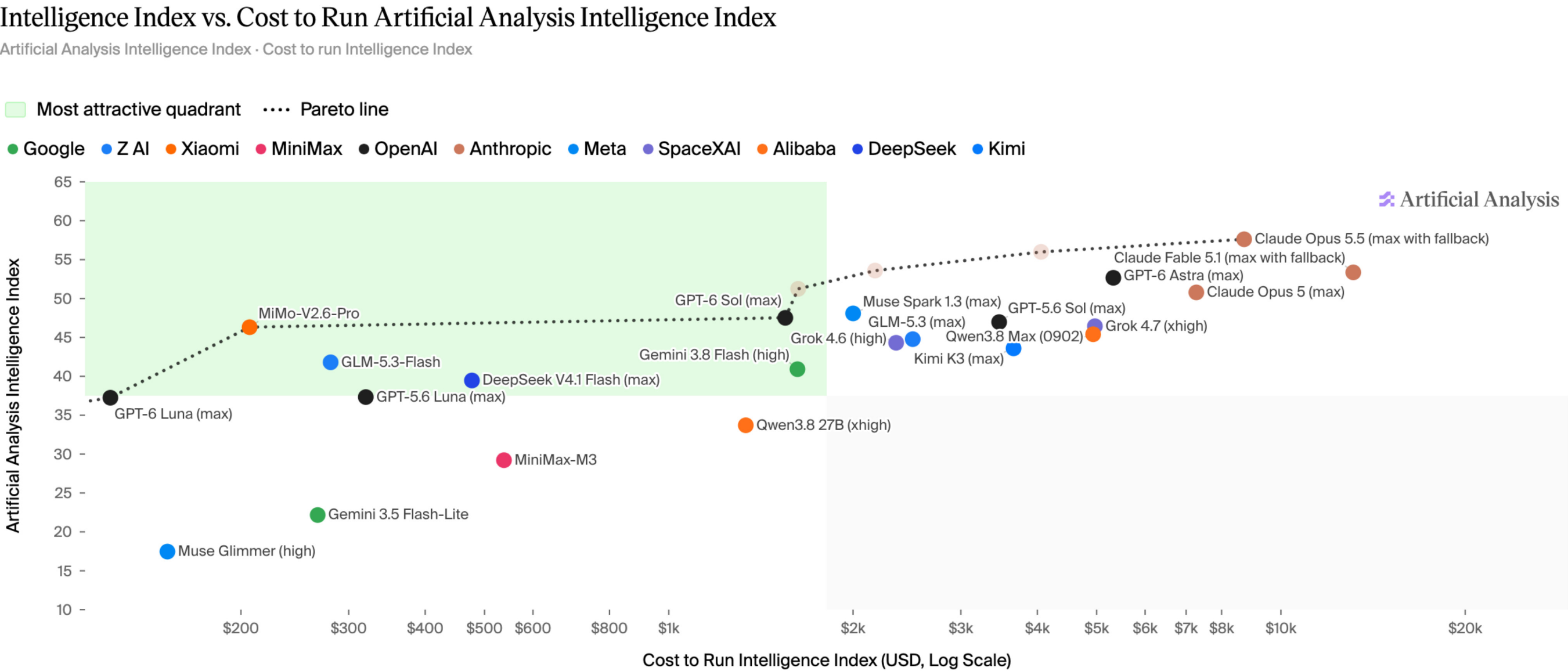
Artificial Analysis



A subscription covers daily agent use for a flat monthly fee.

Source: [Artificial Analysis](#), 22 Sep 2026

How much capability each dollar buys



More capability costs more. The green area is the best value.

Source: [Artificial Analysis](#), 23 Sep 2026

When the laptop is not enough

Where does each one run?

the agent session

inference

the experiment code

- **Heavy work runs on compute nodes**, with your approval.
- Instructions do not enforce limits. **Permissions and quotas do.**
- Know what reaches the model provider.

● LIVE

Submit the job that
regenerates today's
embeddings for 5,991
proteins.

STRONGLY RECOMMENDED

Use Git for any real work. *You hold the record.*

 **Undo anything**

 **See every change**

 **Experiment safely**

 **Own the result**

 **Defend every claim**

 **Share and reproduce**

HOW WE USE IT TODAY



Two skills: before and after

own-it

BEFORE ANYTHING RUNS

- Interviews you: question, data, metric, baseline, budget.
- Ends in `PLAN.md` . You commit it.

It shows how many decisions the model would make without you.

Which split, which metric, what counts as better.

sign-it

BEFORE YOU SUBMIT

- Up to ten claims and decisions you would have to defend.

accept

qualify

reject

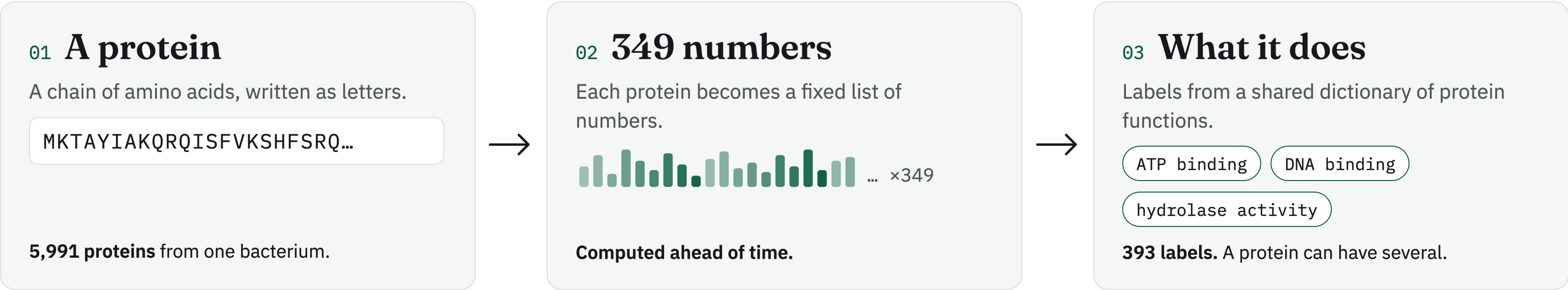
don't know

- Not signing is a valid outcome.

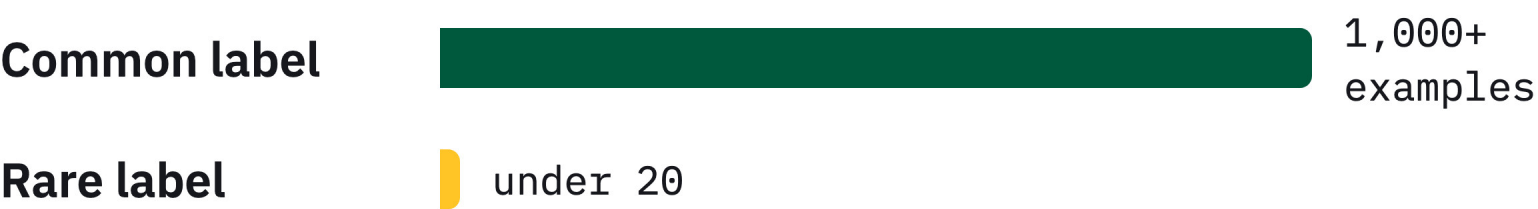
More research skills: github.com/imbad0202/academic-research-skills

BACKGROUND, IN PLAIN WORDS

What the model is trying to learn



WHY IT IS HARD



A model that treats every example alike learns rare labels as “almost never.”

Today’s question: does giving rare examples more weight help, *and how would you know?*

Today’s task

“Class imbalance handling is completely missing.”

– a hypothetical reviewer

THE ONE CHANGE

Balanced class weights, so each term’s positives and negatives count equally.

go-imbalance-sprint

5,991

proteins

393

MF terms

349

precomputed features

REPOSITORY

BASELINE

Logistic regression, unweighted.

0.728

micro F1 on test

0.52

rare-term F1

Decide before you run:

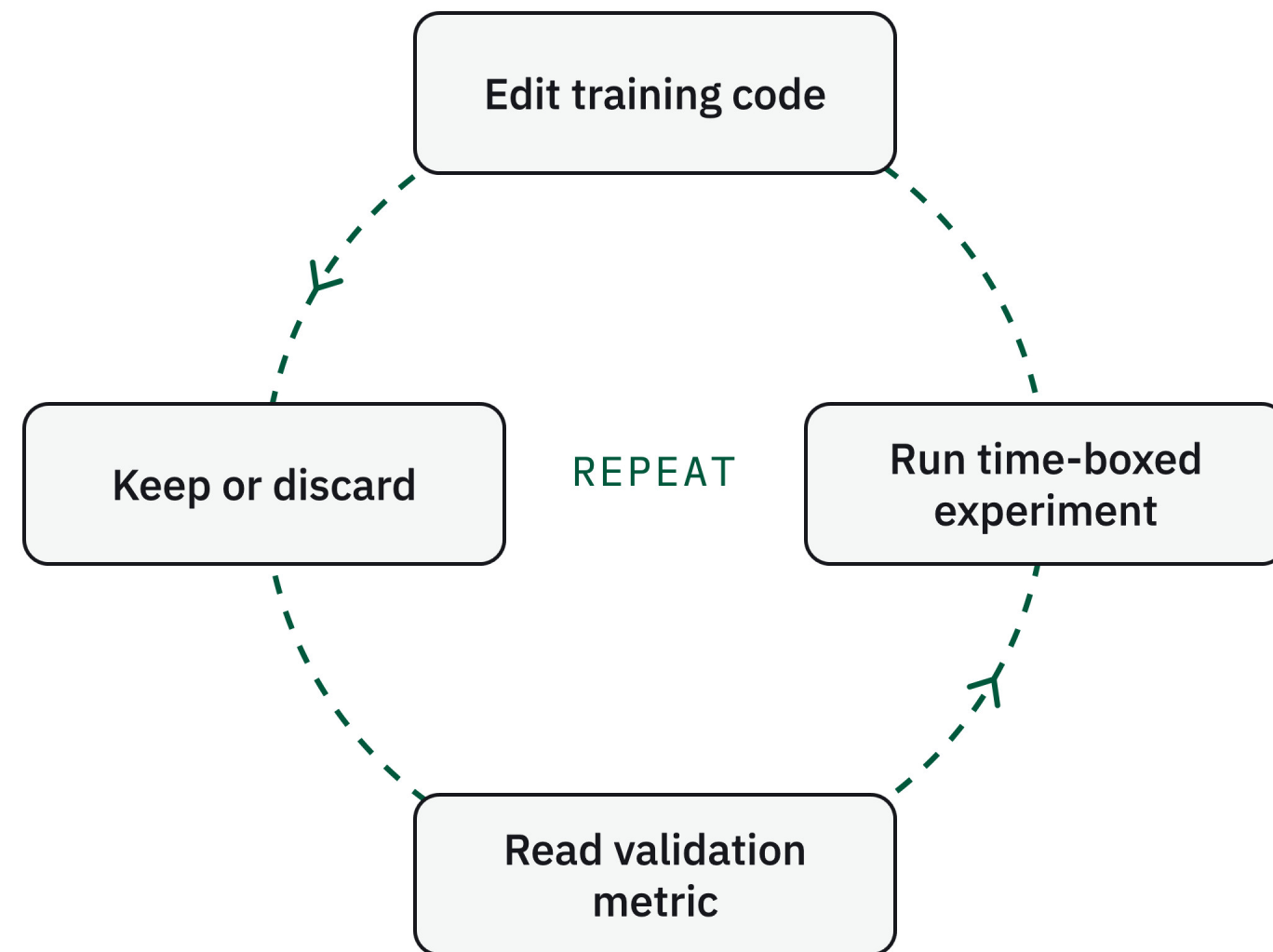
Micro F1

Macro F1

Rare-term F1

They will not agree.

From one experiment to a loop



Karpathy's autoresearch runs this loop on its own.

THE RESEARCHER SETS

objective

evaluation

editable scope

budget

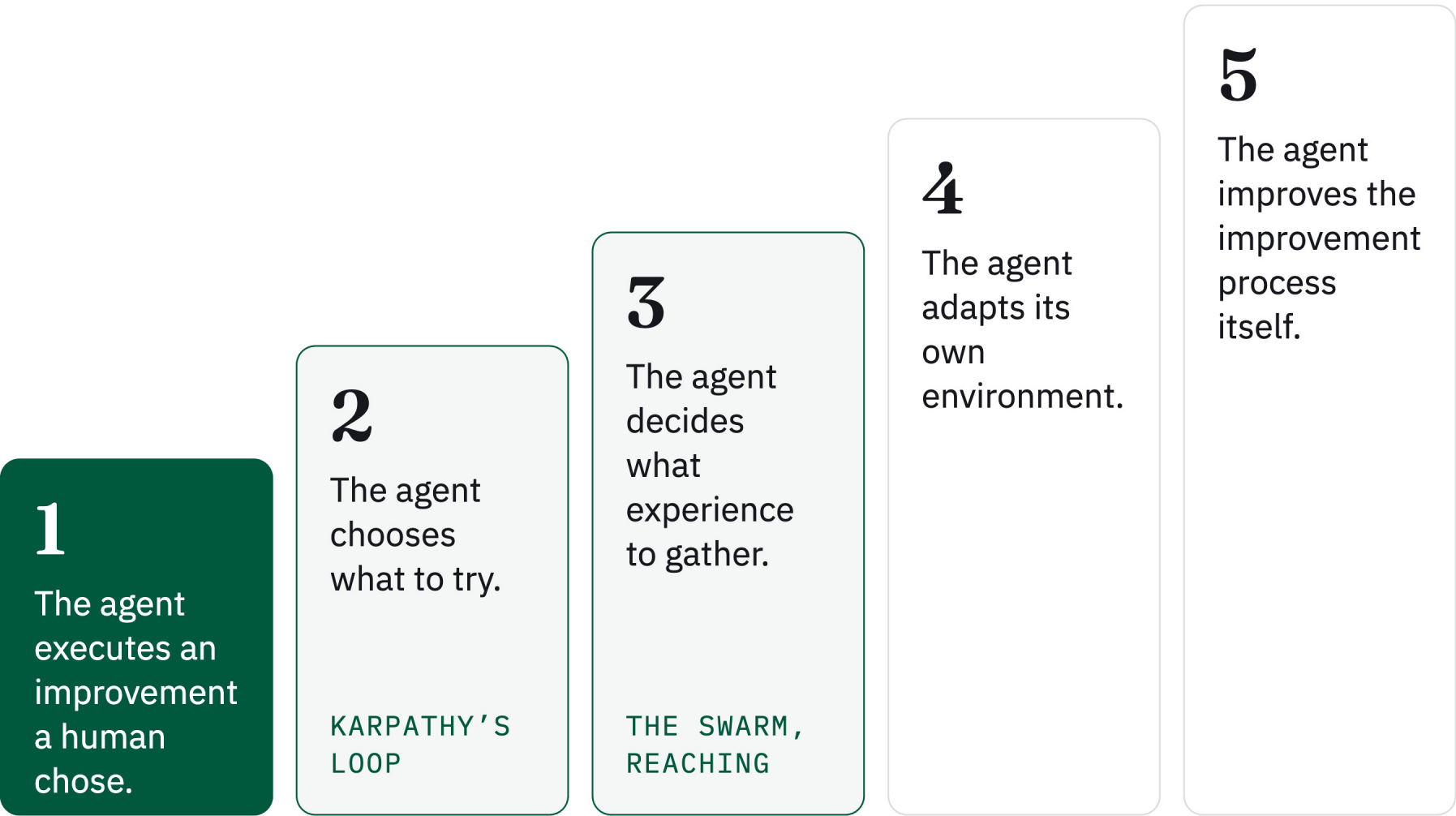
stopping rule

Optimizing a benchmark is *not* establishing a finding.

Where this is going

A LADDER OF AUTONOMY

Duan et al., *The Last AI Built by Humans*



SIGNAL

A swarm of agents beat the state of the art on NanoChat in three days. Kondratyuk, Sep 2026

THE RECORD IS THE ENGINE

Past attempts become training data for the next round.
Dream-RSI

The scale is different. *The need for a signer is not.*

Hands-on

1 Open your harness in `go-imbalance-sprint`.

2 **Ask it: “which skills do you have?”**
It should name own-it and sign-it. If not, paste `skills/own-it/
SKILL.md`.

3 Say “*own it.*”

SPRINT CLOCK

40:00